

TRANSPORT PHENOMENA II, CHE 312

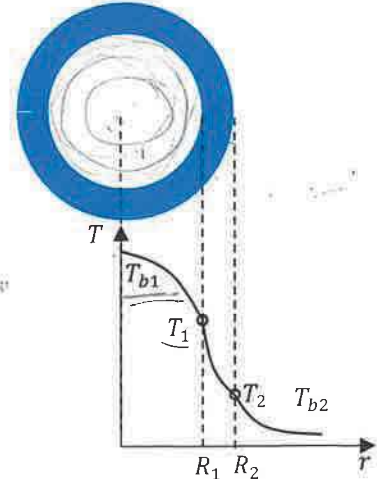
EXAM 3

CLOSED BOOK, TIME: 120 MINUTES

1. Consider a cylindrical pipe with an inner radius R_1 , an outer radius R_2 , and a constant thermal conductivity k .

- 1.1. Using the equation of energy in cylindrical coordinates, derive an equation for the steady-state temperature profile and the heat flux on the inner surface of the pipe in terms of T_1 and T_2 , the temperatures on the inner and outer surface of the pipe.

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi$$



- 1.2. A fluid with bulk temperature T_{b1} and heat transfer coefficient h_1 is in contact with the inner surface of the pipe while a second fluid with bulk temperature T_{b2} and heat transfer coefficient h_2 is in contact with the outer surface of the pipe. Derive an equation for the overall heat transfer coefficient U_1 for heat exchange between the two fluids with reference to the inner radius R_1 .

Hint: Heat flux on the inner surface is $q_r|_{r=R_1} = h_1(T_{b1} - T_1) = U_1(T_{b1} - T_{b2})$ and the one on the outer surface is $q_r|_{r=R_2} = h_2(T_2 - T_{b2}) = \frac{R_1}{R_2} q_r|_{r=R_1}$.

1.1 $T = T(r) \quad v_z = v_r = v_\theta = 0$

$$0 = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right)$$

$$\Rightarrow \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$BC \begin{cases} r = R_1 \Rightarrow T = T_1 \\ r = R_2 \Rightarrow T = T_2 \end{cases}$$

$$r \frac{dT}{dr} = C_1$$

$$\frac{dT}{dr} = \frac{C_1}{r}$$

$$T = C_1 \ln r + C_2$$

$$T_1 = C_1 \ln R_1 + C_2$$

$$T_2 = C_1 \ln R_2 + C_2$$

$$T_1 - T_2 = C_1 \ln \frac{R_1}{R_2}$$

$$C_1 = \frac{T_1 - T_2}{\ln \frac{R_1}{R_2}}$$

$$C_2 = T_1 - \frac{T_1 - T_2}{\ln \frac{R_1}{R_2}} \ln R_1$$

$$T = \frac{T_1 - T_2}{\ln \frac{R_1}{R_2}} \ln r + T_1 - \frac{T_1 - T_2}{\ln \frac{R_1}{R_2}} \ln R_1$$

$$T = T_1 + \frac{T_1 - T_2}{\ln \frac{R_1}{R_2}} \ln \frac{r}{R_1}$$

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2.1

$$N_{Ay} = x_A (N_{Ay} + N_{Dy}) - c D_{AD} \frac{dx_A}{dy}$$

$$= -c D_{AB} \frac{dx_A}{dy}$$

$$= -D_{AB} \frac{dc_A}{dy}$$

$$c_A = c_A(y, t)$$

Shell

$$WL \int_{-b}^b \frac{\partial c_A}{\partial t} dy = WL N_{Ay}|_y - WL N_{Ay}|_{y+dy}$$

$$\frac{\partial c_A}{\partial t} = + D_{AB} \frac{\partial^2 c_A}{\partial y^2}$$

$$\frac{\partial c_A}{\partial t} = - \frac{D_{AB}}{b^2} \frac{\partial^2 c_A}{\partial y^2}$$

$$- \frac{D_{AB} c_{A0}}{b^2} \frac{\partial \theta}{\partial \tau} = - \frac{D_{AB}}{b^2} + \frac{c_{A0}}{b^2} \frac{\partial^2 \theta}{\partial n^2}$$

$$- \frac{D_{AB} c_{A0}}{b^2} \frac{\partial \theta}{\partial \tau} = - \frac{D_{AB} c_{A0}}{b^2} \frac{\partial^2 \theta}{\partial n^2}$$

$$\frac{\partial^2 \theta}{\partial n^2} = \frac{\partial \theta}{\partial \tau}$$

$$BC \begin{cases} \theta(n=\pm 1, \tau > 0) = 0 \\ \theta(n, \tau=0) = 1 \end{cases}$$

$$BC \begin{cases} c_A(y=\pm b, t > 0) = c_{A0} \\ c_A(y, t=0) = 0 \end{cases} \quad \theta = \theta(n, \tau)$$

$$\theta = \frac{c_{A0} - c_A}{c_{A0}} \quad \frac{\partial c_A}{\partial t} = \frac{\partial c_A}{\partial \theta} \cdot \frac{\partial \theta}{\partial \tau} \cdot \frac{\partial \tau}{\partial t}$$

$$\tau = \frac{D_{AB} t}{b^2} \quad n = \frac{y}{b} \quad = -c_{A0} \cdot \frac{\partial \theta}{\partial \tau} \cdot \frac{D_{AB}}{b^2}$$

$$c_A = c_{A0} (1 - \theta) = c_{A0} - c_{A0} \theta = - \frac{D_{AB} c_{A0}}{b^2} \frac{\partial \theta}{\partial \tau}$$

$$f(n) = \frac{\partial c_A}{\partial y} = \frac{\partial c_A}{\partial \theta} \cdot \frac{\partial \theta}{\partial n} \cdot \frac{\partial n}{\partial y} = -c_{A0} \cdot \frac{\partial \theta}{\partial n} \cdot \frac{1}{b} = - \frac{c_{A0}}{b} \cdot \frac{\partial \theta}{\partial n}$$

$$\frac{\partial^2 c_A}{\partial y^2} = \frac{\partial}{\partial y} f(n) = \frac{\partial f}{\partial n} \cdot \frac{\partial n}{\partial y} = - \frac{c_{A0}}{b} \frac{\partial^2 \theta}{\partial n^2} \cdot \frac{1}{b}$$

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